CS214 Final Project Design Document 1

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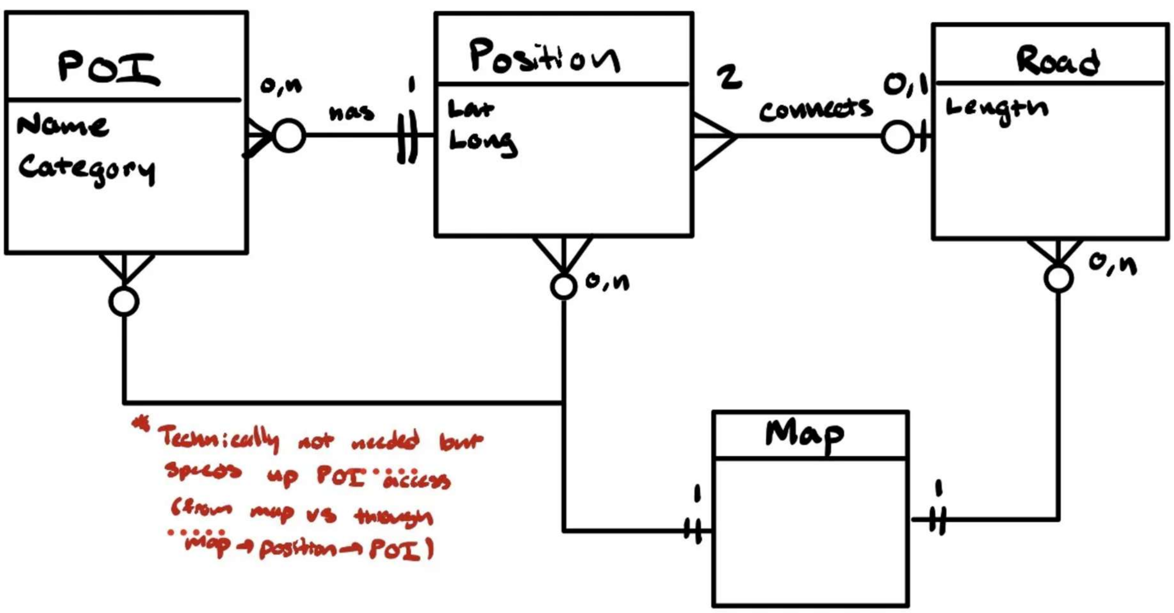
1. ERD

Entities (with attributes) are displayed below. Relationships between entities are described using Crow's Feet Notation.



2. Building Blocks and ADTs

Entity Representation

<u>Map</u>: Weighted Undirected Graph (further described below). WU Graph because roads have a length and do not have an associated direction (two-way).

<u>POI:</u> Struct with name and category

<u>Position:</u> Struct with latitude, longitude (associated POIs described below)

<u>Road</u>: Struct with position1, position2, length

Data Structures

Collection of Roads: Represented in Map: ***edges*** in the WU graph are roads, the ***weights*** of edges correspond to length of roads

Collection of Positions: Represented in Map: each ***vertex*** of the WU graph will be a position. Needs a ***dictionary*** to map position coordinates (lat, long) to an integer (vertex). Use a second to map vertices to position coordinates as well.

Collection of POI(s) *(by position):* Take advantage of the graph representation of the Map (with integer vertices) to have direct access to POIs instead of top-down linear access through positions. Use a ***dictionary*** to map vertex to a **set** of POI structs at the associated position.

Additional data structures to optimize the three queries required:

Collection of POI(s) *(by category):* Use a ***dictionary*** to map a category to a **set** of POIs in the associated category.

Collection of POI Locations *(by name)*: Use a ***dictionary*** to map a POI name to the *position* of a POI (i.e. the vertex on the graph representing the position).